Ex - 07
Data Visualization Using Power BI

AIM

To visualize data using Power BI by connecting data sources, creating charts, applying DAX calculations, and building an interactive dashboard using the Cardio dataset.

PROBLEM STATEMENT

To analyze the Cardio dataset using Power BI by creating bar charts, line charts, pie charts, calculated columns, and measures to present meaningful insights such as cardio cases by age group, cholesterol levels, and blood pressure trends.

PROCEDURE

1. Open Microsoft Power BI Desktop and explore the Report View, Data View, Model View, and Visualizations pane.
2. Connect to the Cardio dataset (CSV) using Home > Get Data, select the required file, and click Load to import the data.
3. Connect to a SQL database by selecting Home > Get Data > SQL Server, entering the server details, and selecting the required tables.
4. Open Power Query Editor using Transform Data to remove duplicates, handle missing values, rename columns, and change data types.
5. Create a bar chart by selecting the Clustered Bar Chart visual, dragging an age group field to the Y-axis and the count of id to the X-axis, and filtering cardio = 1 to show cardio cases by age group.
6. Create a line chart by selecting the Line Chart visual, placing age on the X-axis and the average of ap_hi (systolic blood pressure) on the Y-axis to analyze the trend.
7. Create a pie chart by selecting the Pie Chart visual, placing cardio in the Legend and the count of id in Values to display the distribution of patients with and without cardiovascular disease.
8. Create a calculated column using Table tools > New column and enter a DAX formula to calculate additional values for each row.
9. Create measures using Table tools > New measure and apply DAX functions such as SUM(), COUNTROWS(), DISTINCTCOUNT() and AVERAGE() to calculate totals, counts and averages.
10. Build an interactive dashboard-style report by arranging charts, KPI cards, and slicers, formatting titles and colors, testing filters, and saving the report as a .pbix file.

IMPLEMENTATION

1. Calculated Column (DAX)

```
Age Group = IF(Cardio[age] >= 50, "50+", "Below 50")
```

2. Calculated Column (DAX)

```
BMI = DIVIDE(Cardio[weight], POWER(Cardio[height] / 100, 2))
```

3. Measure - Total Patients

```
Total Patients = COUNTROWS(Cardio)
```

4. Measure - Average Age

```
Average Age = AVERAGE(Cardio[age])
```

5. Measure - Cardio Cases

```
Cardio Cases = CALCULATE(COUNTROWS(Cardio), Cardio[cardio] = 1)
```

Dashboard components:

• Cards - Total Patients, Average Age, Cardio Cases

• Bar chart - Cardio cases by age group

• Column chart - Patients by cholesterol level

• Line chart - Average systolic BP by age

• Pie chart - Cardio disease distribution

• Slicer - Filter by gender or cholesterol

Note: Replace the table and column names with those in your dataset.

OUTPUT



Fig: Cardio Analytics Dashboard (Power BI)

RESULT

Data was successfully visualized using Power BI on the Cardio dataset. Bar charts, line charts, pie charts, DAX calculated columns, and measures were created and combined into an interactive dashboard for effective data analysis.